

ACADEMIC POSITIONS

University of Notre Dame du Lac

Society of Science Postdoctoral Fellow.

Indiana, USA

2025 - 2028

Instituto de Ciencias Matemáticas

Predoctoral Researcher for the Spanish National Research Council.

Ph.D. in Geometry and Algebra.

Madrid, Spain

2021 - 2025

- Advisor: Prof. Javier Aramayona
- Research area: Mapping class groups & Teichmüller theory

EDUCATION

Universidad Autónoma de Madrid

Master in Mathematics and Applications

Madrid, Spain

2020 - 2021

- Master thesis: Modular groups and arc complexes **(10/10)**
- GPA: 9.64/10

Universidad Autónoma de Madrid

Double degree in Mathematics and Computer Science

Madrid, Spain

2015 - 2020

- Final year project for Mathematics: Classification of surfaces **(10/10)**
- Awarded 29 *Matrículas de Honor* (Highest distinction).
- GPA: 9.30/10

PUBLICATIONS

1. Homomorphisms between pure mapping class groups.
Advances in Mathematics (2025).
2. Non-planar ends are continuously unforgettable.
Joint with J. Aramayona, R. Skipper, J. Tao, N. G. Vlamis and X. Wu.
International Mathematics Research Notices (2025).
3. Braided multitwists.
Algebraic and Geometric Topology (2025).
4. Finite rigid sets of the non-separating curve complex.
Forum Mathematicum (2022).
5. Matching arc complexes: Connectedness and hyperbolicity.
Joint with J. Aramayona and A. Fernández.
Expositiones Mathematicae (2021).
6. Towards the Automatic and Schedule-Aware Alerting of Internetwork Time Series.
Joint with D. Perdices, J. García-Dorado, J. Ramos, J. Aracil.
IEEE Access (2021).

PREPRINTS

1. Rigidity of maps between configuration spaces.
Joint with P. Huxford, D. Minahan and J. Schillewaert.
In arXiv:2607.05826 (2026).
2. Holomorphic maps between moduli spaces II.
Joint with J. Souto.
In arXiv:2412.01257 (2024).

RESEARCH STAYS

Institut de Recherche Mathématique de Rennes

Collaboration with prof. Juan Souto

Rennes, France

April - June 2024

Institut Henri Poincaré

Scientific Program:

Groups acting on Fractals, Hyperbolicity and Self-Similarity

Paris, France

April - June 2022

AWARDS, FELLOWSHIPS, & GRANTS

Provost Society of Science Fellowship

University of Notre Dame du Lac.

2025 - 2028

FPI predoctoral contract

Spanish Ministry of Science and Innovation.

2021 - 2025

JAE Intro SOMdM Grant

Research grant awarded by Spanish National Research Council.

2020 - 2021

Excellence scholarships.

Awarded yearly by the Community of Madrid.

2015 - 2020

INVITED TALKS, SEMINARS & COLLOQUIA

Wasatch Topology Conference

Polynomials, configurations and braids.

August 19-21, 2026

Redbud Topology Conference

Rigidity of moduli spaces and mapping class groups.

April 11, 2026

University of Notre Dame

Rigidity results for the mapping class group.

September 23, 2025

Instituto de Ciencias Matemáticas

Resultados de rigidez del grupo modular.

May 22, 2025

Universidad Nacional Autónoma de México

Braided elements & homomorphisms in the mapping class group.

October 28, 2024

Universidad Complutense de Madrid

Homomorphisms between mapping class groups.

September 23, 2024

Internationales Wissenschaftsforum Heidelberg <i>Transverse measures to infinite type laminations.</i>	July 8, 2024
Institut de Recherche Mathématique de Rennes <i>Holomorphic rigidity of moduli space.</i>	June 24, 2024
Institut de Recherche Mathématique de Rennes <i>Mapping class groups, braided elements and homomorphisms.</i>	June 19, 2024
Instituto de Ciencias Matemáticas <i>Braided multitwists in the mapping class group.</i>	December 5, 2023
Universidad de León <i>Finite rigidity of non-separating curve complexes.</i>	February 9, 2023
Universidad Autónoma de Madrid <i>Introduction to geometric group theory.</i>	December 14, 2022
Instituto de Ciencias Matemáticas <i>The symplectic representation of the mapping class group.</i>	December 2, 2022
Institut Henri Poincaré <i>Finite rigid sets of the non-separating curve complex.</i>	June 2022
Instituto de Ciencias Matemáticas <i>The curve complex.</i>	May 24, 2022
Instituto de Ciencias Matemáticas <i>On the hyperbolicity of arc complexes and some of their relatives.</i>	December 9, 2021

TEACHING EXPERIENCE

Instructor for Calculus B <i>University of Notre Dame du Lac.</i>	Spring Term 25/26
Teaching Assistant for Mathematical Analysis <i>Mathematics, Universidad Autónoma de Madrid.</i>	1st Term 24/25
Teaching Assistant for Linear Algebra <i>Data Science and Engineering, Universidad Autónoma de Madrid.</i>	1st Term 23/24
Teaching Assistant for Mathematics <i>Environmental Science, Universidad Autónoma de Madrid.</i>	1st Term 22/23
Teacher in the ‘Pequeño instituto matemático’ (PIM) <i>PIM is a weekly activity organized by the ICMAT where high-school students discuss math Olympiad style problems.</i>	2022 - 2024

CONFERENCE ORGANIZATION

Geometric Group Theory and Low-dimensional Topology <i>Co-organizar of Graduate School at Instituto de Ciencias Matemáticas.</i>	May 2022
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WORKSHOP & CONFERENCE PARTICIPATION

Wasatch Topology Conference <i>University of Utah and Brigham Young University.</i>	August 2026 <i>Utah, USA</i>
Advanced School in Geometric Group Theory Lluís Santaló <i>Universidad Internacional Menéndez Pelayo.</i>	July 2026 <i>Santander, Spain</i>
Advanced school: Aspherical 2-complexes and their fundamental groups <i>Instituto de Ciencias Matemáticas.</i>	June 2026 <i>Madrid, Spain</i>
Young Geometric Group Theory XIV <i>Korea Institute for Advanced Study.</i>	June 2026 <i>Seoul, South Korea</i>
Curve counting: What's next? <i>Hamilton Mathematics Institute.</i>	May 2026 <i>Dublin, Ireland</i>
Redbud Topology Conference <i>University of Oklahoma.</i>	April 2026 <i>Oklahoma, USA</i>
A conference on Geometry, Topology and Dynamics of Infinite-type surfaces <i>Casa Matemática Oaxaca.</i>	June 2025 <i>Oaxaca, Mexico</i>
Young Geometric Group Theory XIII <i>University of Copenhagen.</i>	April 2025 <i>Copenhagen, Denmark</i>
Group, Topology and Applications VI <i>University of the Basque Country.</i>	April 2025 <i>Bilbao, Spain</i>
Red temática de geometría y física <i>Instituto de Ciencias Matemáticas.</i>	November 2024 <i>Madrid, Spain</i>
Groups in Madrid <i>Instituto de Ciencias Matemáticas.</i>	October 2024 <i>Madrid, Spain</i>
Geometry and dynamics of surfaces beyond finite type <i>Heidelberg University.</i>	July 2024 <i>Heidelberg, Germany</i>
Young Geometric Group Theory XII <i>University of Bristol.</i>	April 2024 <i>Bristol, UK</i>
Groups in Madrid <i>Instituto de Ciencias Matemáticas.</i>	October 2023 <i>Madrid, Spain</i>
Group Actions and Low-dimensional Topology <i>Organized by Instituto de Ciencias Matemáticas.</i>	July 2023 <i>Barco de Ávila, Spain</i>
Groups and Dynamics in Geometry <i>Organized by ETH Zürich.</i>	June 2023 <i>Ascona, Switzerland</i>
Young Geometric Group Theory XI <i>University of Münster.</i>	February 2023 <i>Münster, Germany</i>
Young researchers congress of the Real Sociedad Matemática Española <i>Universidad de León.</i>	February 2023 <i>León, Spain</i>

Groups, Topology and Applications IV

University of the Basque Country.

February 2023

Bilbao, Spain

Groups in Madrid

Instituto de Ciencias Matemáticas.

October 2022

Madrid, Spain

Groups acting on Fractals

Centre International de Rencontres Mathématiques.

April 2022

Luminy, France

Biannual meeting of the Real Sociedad Matemática Española

University of Castilla-La Mancha

January 2022

Ciudad Real, Spain

Groups in Madrid

Instituto de Ciencias Matemáticas.

November 2021

Madrid, Spain